

Doğukan Rıza Demircan

2025-11-26

```
library(readxl)

goc_new <- read_xls("vatandaslik ulkesine gore turkiye_ye gelen ve turkiye_den giden goc (2).xls")

## New names:
## * `` -> `...2`
## * `` -> `...3`
## * `` -> `...4`
## * `` -> `...5`
## * `` -> `...6`
## * `` -> `...7`
## * `` -> `...8`
## * `` -> `...9`
## * `` -> `...10`

install.packages("dplyr")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)

library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

goc_new <- read_excel(
  "vatandaslik ulkesine gore turkiye_ye gelen ve turkiye_den giden goc (2).xls",
  skip = 4
)

## New names:
## * `` -> `...1`
## * `` -> `...2`
## * `Toplam Total` -> `Toplam Total...3`
## * `Erkek Male` -> `Erkek Male...4`
## * `Kadın Female` -> `Kadın Female...5`
## * `` -> `...6`
## * `Toplam Total` -> `Toplam Total...7`
## * `Erkek Male` -> `Erkek Male...8`
## * `Kadın Female` -> `Kadın Female...9`
```

```

## * `` -> `...10`
goc_new <- goc_new %>%
  slice_head(n = nrow(goc_new) - 10)

goc_new <- goc_new[, -6]

goc_new$...1[1:108] <- 2024

goc_new$...1[109:212] <- 2023

goc_new$...1[213:326] <- 2022

goc_new$...1[327:456] <- 2021

goc_new$...1[457:666] <- 2020

goc_new <- goc_new[1:666, ]

install.packages("ggplot2")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
library(ggplot2)

library(dplyr)
install.packages("tidyr")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
library(tidyr)

goc_new <- goc_new %>%
  rename(ülkeler = ...2)
colnames(goc_new)[3] <- "gelen_goc"
colnames(goc_new)[9] <- "toplamlam"
colnames(goc_new)[6] <- "giden_goc"

install.packages("treemapify")

## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
library(treemapify)

turk_devletleri <- c(
  "Azerbaycan-Azerbaijan",
  "Türkmenistan-Turkmenistan",
  "Özbekistan-Uzbekistan",
  "Kazakistan-Kazakhstan",
  "Kırgızistan-Kyrgyzstan"
)

turk <- goc_new %>%
  filter(ülkeler %in% turk_devletleri,
    ...1 %in% 2020:2024) %>%
  select(Ulke = ülkeler, ...1, Toplam = gelen_goc)

```

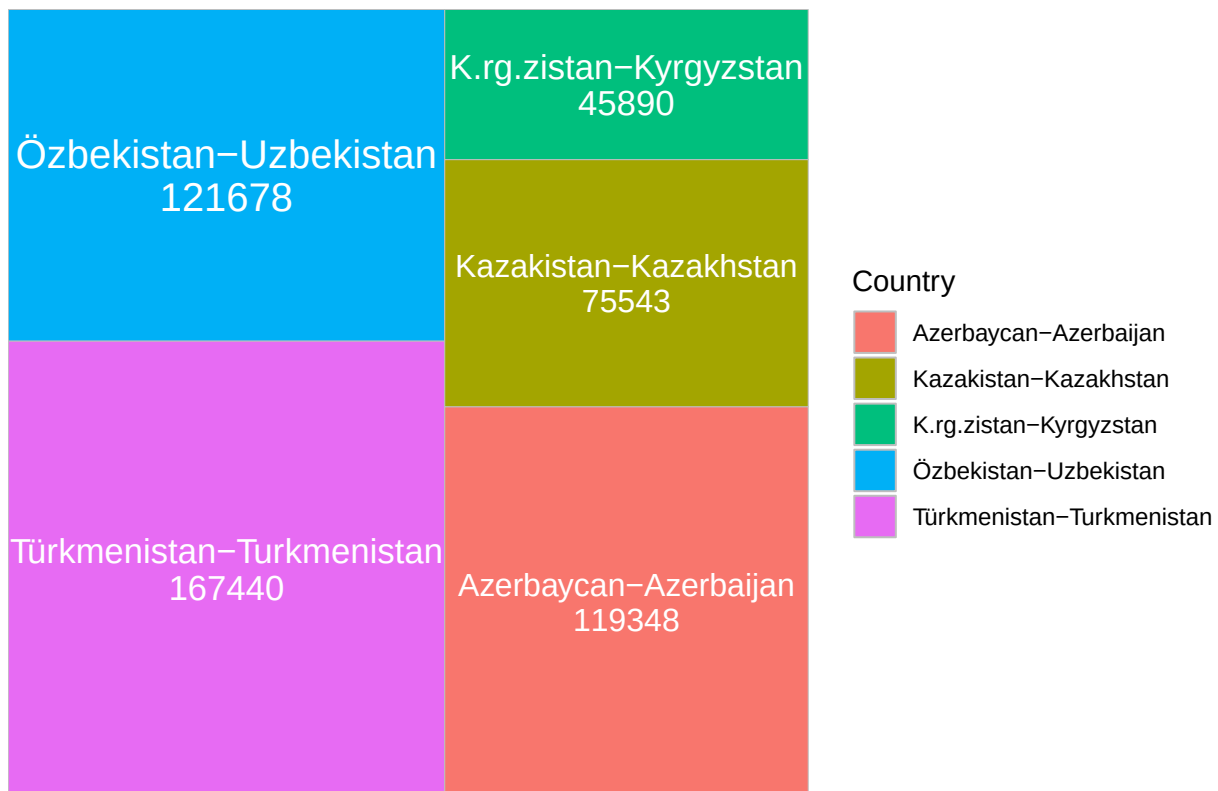
```

turk_giden_toplam <- turk %>%
  group_by(Ulke) %>%
  summarise(Toplam_2020_2024 = sum(Toplam, na.rm = TRUE))

ggplot(turk_giden_toplam, aes(area = Toplam_2020_2024,
  fill = Ulke,
  label = paste0(Ulke, "\n", Toplam_2020_2024))) +
  geom_treemap() +
  geom_treemap_text(colour = "white", size = 15, place = "centre") +
  labs(
    title = "2020-2024 Turkic States Arriving in Türkiye",
    fill = "Country"
  ) +
  theme_minimal()

```

2020–2024 Turkic States Arriving in Türkiye



#This graph shows citizens of other Turkic states who emigrated from their home countries to Turkey between 2020 and 2024. During this period, the highest number of migrants to Turkey were Uzbekistani citizens, while the lowest number were Kyrgyz citizens.

```

turk_devletleri <- c(
  "Azerbaycan-Azerbaijan",
  "Türkmenistan-Turkmenistan",
  "Özbekistan-Uzbekistan",
  "Kazakistan-Kazakhstan",
  "Kırgızistan-Kyrgyzstan"
)

```

```

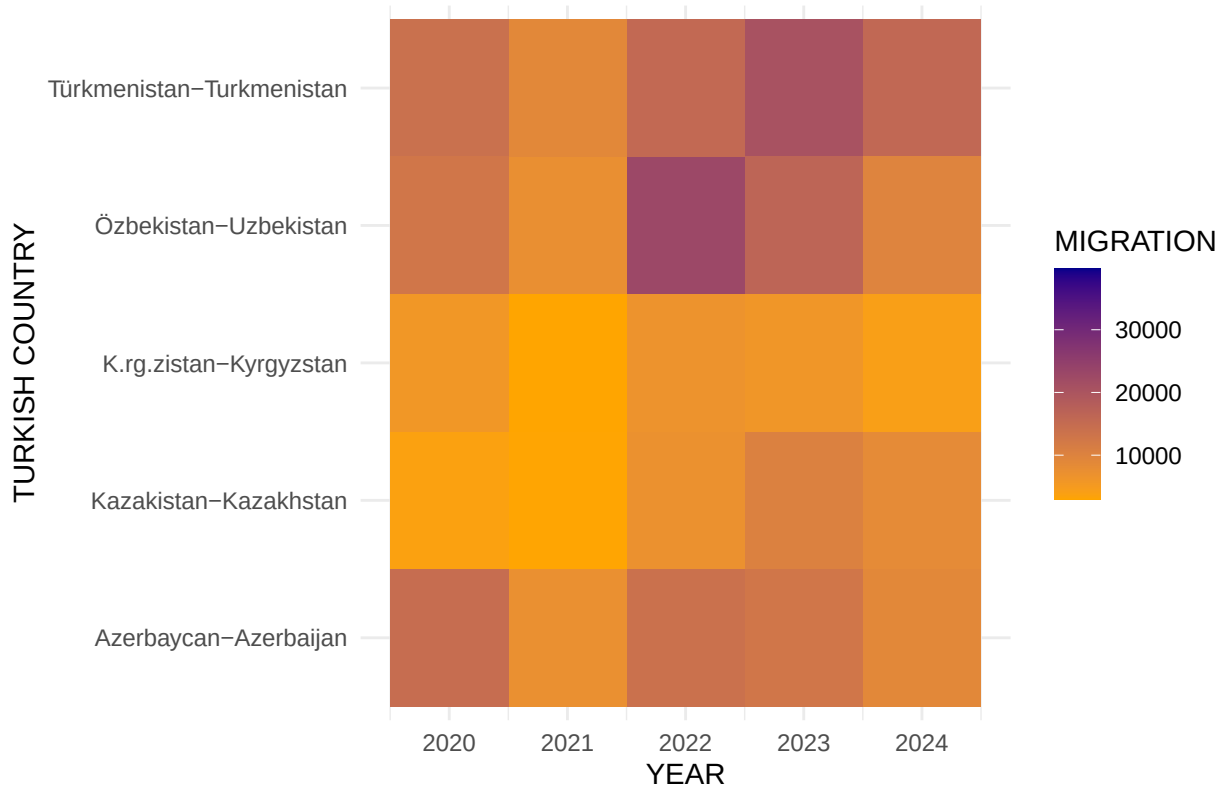
giden_goc1 <- goc_new %>%
  filter(ülkeler %in% turk_devletleri,
    ...1 %in% 2020:2024) %>%
  select(Ulke = ülkeler, Yil = ...1, Giden = giden_goc
  )

giden_goc1$Giden <- as.numeric(giden_goc1$Giden)
giden_goc1$Yil <- as.integer(trimws(giden_goc1$Yil))

ggplot(giden_goc1, aes(x = Yil, y = Ulke, fill = Giden)) +
  geom_tile() +
  scale_fill_gradient(low = "orange", high = "darkblue") +
  labs(title = "Heat Map of Migration Density from Türkiye to Their Home Countries",
    x = "YEAR", y = "TURKISH COUNTRY", fill = "MIGRATION") +
  theme_minimal()+
  theme(
    plot.title = element_text(hjust = 0, size = 14, face = "bold"),
    plot.title.position = "plot"
  )

```

Heat Map of Migration Density from Türkiye to Their Home Countries



#This chart shows the citizens of Kyrgyzstan, Uzbekistan, Azerbaijan, and Kazakhstan who returned from Türkiye between 2020 and 2024. Turkmenistan and Azerbaijan saw the most migration in 2020, followed by Turkmenistan in 2021, 2023, and 2024, and Uzbekistan in 2022.

```

turk_devletleri <- c(
  "Azerbaycan-Azerbaijan",
  "Türkmenistan-Turkmenistan",
  "Özbekistan-Uzbekistan",
  "Kazakistan-Kazakhstan",

```

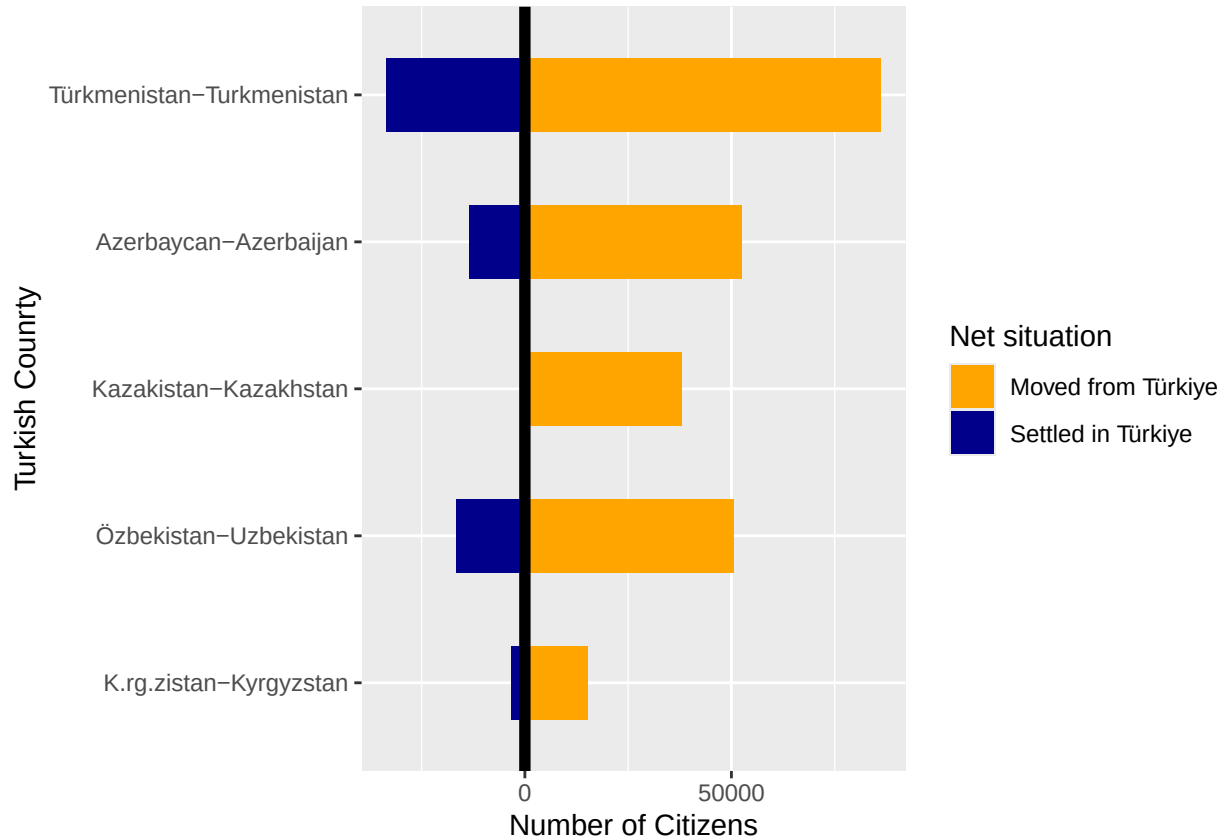
```

"Kırgızistan-Kyrgyzstan"
)
toplamlar_1 <- gdc_new %>%
  filter(ülkeler %in% türk_devletleri,
    ...1 %in% 2020:2024) %>%
  select(Ülke = ülkeler, Yıl = ...1, toplam = toplam)

toplamlar_1$Giden <- as.numeric(toplamlar_1$toplam)
toplamlar_1$Yıl <- as.integer(trimws(toplamlar_1$Yıl))

ggplot(toplamlar_1, aes(x = reorder(Ülke, toplam),
  y = toplam,
  fill = toplam < 0)) +
  geom_col(width = 0.5) +
  geom_hline(yintercept = 0, color = "black", linewidth = 2) +
  scale_fill_manual(values = c("orange", "darkblue"),
    labels = c("Moved from Türkiye", "Settled in Türkiye"),
    name = "Net situation") +
  coord_flip() +
  labs(
    x = "Turkish Country",
    y = "Number of Citizens")

```



```

theme_minimal()

## <theme> List of 144
## $ line : <ggplot2::element_line>
## ..@ colour : chr "black"

```

```

## ..@ linewidth      : num 0.5
## ..@ linetype       : num 1
## ..@ lineend        : chr "butt"
## ..@ linejoin       : chr "round"
## ..@ arrow          : logi FALSE
## ..@ arrow.fill     : chr "black"
## ..@ inherit.blank  : logi TRUE
## $ rect              : <ggplot2::element_rect>
## ..@ fill           : chr "white"
## ..@ colour         : chr "black"
## ..@ linewidth      : num 0.5
## ..@ linetype       : num 1
## ..@ linejoin       : chr "round"
## ..@ inherit.blank  : logi TRUE
## $ text              : <ggplot2::element_text>
## ..@ family         : chr ""
## ..@ face           : chr "plain"
## ..@ italic         : chr NA
## ..@ fontweight     : num NA
## ..@ fontwidth      : num NA
## ..@ colour         : chr "black"
## ..@ size           : num 11
## ..@ hjust          : num 0.5
## ..@ vjust          : num 0.5
## ..@ angle          : num 0
## ..@ lineheight     : num 0.9
## ..@ margin         : <ggplot2::margin> num [1:4] 0 0 0 0
## ..@ debug          : logi FALSE
## ..@ inherit.blank  : logi TRUE
## $ title             : <ggplot2::element_text>
## ..@ family         : NULL
## ..@ face           : NULL
## ..@ italic         : chr NA
## ..@ fontweight     : num NA
## ..@ fontwidth      : num NA
## ..@ colour         : NULL
## ..@ size           : NULL
## ..@ hjust          : NULL
## ..@ vjust          : NULL
## ..@ angle          : NULL
## ..@ lineheight     : NULL
## ..@ margin         : NULL
## ..@ debug          : NULL
## ..@ inherit.blank  : logi TRUE
## $ point             : <ggplot2::element_point>
## ..@ colour         : chr "black"
## ..@ shape          : num 19
## ..@ size           : num 1.5
## ..@ fill           : chr "white"
## ..@ stroke         : num 0.5
## ..@ inherit.blank  : logi TRUE
## $ polygon           : <ggplot2::element_polygon>
## ..@ fill           : chr "white"
## ..@ colour         : chr "black"

```

```

## ..@ linewidth      : num 0.5
## ..@ linetype       : num 1
## ..@ linejoin       : chr "round"
## ..@ inherit.blank: logi TRUE
## $ geom              : <ggplot2::element_geom>
## ..@ ink            : chr "black"
## ..@ paper          : chr "white"
## ..@ accent         : chr "#3366FF"
## ..@ linewidth      : num 0.5
## ..@ borderwidth    : num 0.5
## ..@ linetype       : int 1
## ..@ bordertype     : int 1
## ..@ family         : chr ""
## ..@ fontsize       : num 3.87
## ..@ pointsize      : num 1.5
## ..@ pointshape     : num 19
## ..@ colour         : NULL
## ..@ fill           : NULL
## $ spacing           : 'simpleUnit' num 5.5points
## ..- attr(*, "unit")= int 8
## $ margins            : <ggplot2::margin> num [1:4] 5.5 5.5 5.5 5.5
## $ aspect.ratio       : NULL
## $ axis.title         : NULL
## $ axis.title.x       : <ggplot2::element_text>
## ..@ family          : NULL
## ..@ face            : NULL
## ..@ italic          : chr NA
## ..@ fontweight      : num NA
## ..@ fontwidth       : num NA
## ..@ colour          : NULL
## ..@ size            : NULL
## ..@ hjust           : NULL
## ..@ vjust           : num 1
## ..@ angle           : NULL
## ..@ lineheight      : NULL
## ..@ margin          : <ggplot2::margin> num [1:4] 2.75 0 0 0
## ..@ debug           : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.title.x.top   : <ggplot2::element_text>
## ..@ family          : NULL
## ..@ face            : NULL
## ..@ italic          : chr NA
## ..@ fontweight      : num NA
## ..@ fontwidth       : num NA
## ..@ colour          : NULL
## ..@ size            : NULL
## ..@ hjust           : NULL
## ..@ vjust           : num 0
## ..@ angle           : NULL
## ..@ lineheight      : NULL
## ..@ margin          : <ggplot2::margin> num [1:4] 0 0 2.75 0
## ..@ debug           : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.title.x.bottom : NULL

```

```

## $ axis.title.y : <ggplot2::element_text>
##   ..@ family      : NULL
##   ..@ face        : NULL
##   ..@ italic      : chr NA
##   ..@ fontweight  : num NA
##   ..@ fontwidth   : num NA
##   ..@ colour      : NULL
##   ..@ size        : NULL
##   ..@ hjust       : NULL
##   ..@ vjust       : num 1
##   ..@ angle       : num 90
##   ..@ lineheight  : NULL
##   ..@ margin      : <ggplot2::margin> num [1:4] 0 2.75 0 0
##   ..@ debug       : NULL
##   ..@ inherit.blank: logi TRUE
## $ axis.title.y.left : NULL
## $ axis.title.y.right : <ggplot2::element_text>
##   ..@ family      : NULL
##   ..@ face        : NULL
##   ..@ italic      : chr NA
##   ..@ fontweight  : num NA
##   ..@ fontwidth   : num NA
##   ..@ colour      : NULL
##   ..@ size        : NULL
##   ..@ hjust       : NULL
##   ..@ vjust       : num 1
##   ..@ angle       : num -90
##   ..@ lineheight  : NULL
##   ..@ margin      : <ggplot2::margin> num [1:4] 0 0 0 2.75
##   ..@ debug       : NULL
##   ..@ inherit.blank: logi TRUE
## $ axis.text : <ggplot2::element_text>
##   ..@ family      : NULL
##   ..@ face        : NULL
##   ..@ italic      : chr NA
##   ..@ fontweight  : num NA
##   ..@ fontwidth   : num NA
##   ..@ colour      : chr "#4D4D4DFF"
##   ..@ size        : 'rel' num 0.8
##   ..@ hjust       : NULL
##   ..@ vjust       : NULL
##   ..@ angle       : NULL
##   ..@ lineheight  : NULL
##   ..@ margin      : NULL
##   ..@ debug       : NULL
##   ..@ inherit.blank: logi TRUE
## $ axis.text.x : <ggplot2::element_text>
##   ..@ family      : NULL
##   ..@ face        : NULL
##   ..@ italic      : chr NA
##   ..@ fontweight  : num NA
##   ..@ fontwidth   : num NA
##   ..@ colour      : NULL
##   ..@ size        : NULL

```



```

## ..@ hjust      : NULL
## ..@ vjust      : num 1
## ..@ angle      : NULL
## ..@ lineheight : NULL
## ..@ margin     : <ggplot2::margin> num [1:4] 2.2 0 0 0
## ..@ debug      : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.x.top      : <ggplot2::element_text>
## ..@ family         : NULL
## ..@ face           : NULL
## ..@ italic         : chr NA
## ..@ fontweight     : num NA
## ..@ fontwidth      : num NA
## ..@ colour         : NULL
## ..@ size           : NULL
## ..@ hjust          : NULL
## ..@ vjust          : NULL
## ..@ angle          : NULL
## ..@ lineheight     : NULL
## ..@ margin         : <ggplot2::margin> num [1:4] 0 0 4.95 0
## ..@ debug          : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.x.bottom   : <ggplot2::element_text>
## ..@ family         : NULL
## ..@ face           : NULL
## ..@ italic         : chr NA
## ..@ fontweight     : num NA
## ..@ fontwidth      : num NA
## ..@ colour         : NULL
## ..@ size           : NULL
## ..@ hjust          : NULL
## ..@ vjust          : NULL
## ..@ angle          : NULL
## ..@ lineheight     : NULL
## ..@ margin         : <ggplot2::margin> num [1:4] 4.95 0 0 0
## ..@ debug          : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.y          : <ggplot2::element_text>
## ..@ family         : NULL
## ..@ face           : NULL
## ..@ italic         : chr NA
## ..@ fontweight     : num NA
## ..@ fontwidth      : num NA
## ..@ colour         : NULL
## ..@ size           : NULL
## ..@ hjust          : num 1
## ..@ vjust          : NULL
## ..@ angle          : NULL
## ..@ lineheight     : NULL
## ..@ margin         : <ggplot2::margin> num [1:4] 0 2.2 0 0
## ..@ debug          : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.y.left     : <ggplot2::element_text>
## ..@ family         : NULL

```

```

## ..@ face          : NULL
## ..@ italic        : chr NA
## ..@ fontweight    : num NA
## ..@ fontwidth     : num NA
## ..@ colour        : NULL
## ..@ size          : NULL
## ..@ hjust         : NULL
## ..@ vjust         : NULL
## ..@ angle         : NULL
## ..@ lineheight    : NULL
## ..@ margin        : <ggplot2::margin> num [1:4] 0 4.95 0 0
## ..@ debug         : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.y.right : <ggplot2::element_text>
## ..@ family        : NULL
## ..@ face          : NULL
## ..@ italic        : chr NA
## ..@ fontweight    : num NA
## ..@ fontwidth     : num NA
## ..@ colour        : NULL
## ..@ size          : NULL
## ..@ hjust         : NULL
## ..@ vjust         : NULL
## ..@ angle         : NULL
## ..@ lineheight    : NULL
## ..@ margin        : <ggplot2::margin> num [1:4] 0 0 0 4.95
## ..@ debug         : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.theta   : NULL
## $ axis.text.r       : <ggplot2::element_text>
## ..@ family        : NULL
## ..@ face          : NULL
## ..@ italic        : chr NA
## ..@ fontweight    : num NA
## ..@ fontwidth     : num NA
## ..@ colour        : NULL
## ..@ size          : NULL
## ..@ hjust         : num 0.5
## ..@ vjust         : NULL
## ..@ angle         : NULL
## ..@ lineheight    : NULL
## ..@ margin        : <ggplot2::margin> num [1:4] 0 2.2 0 2.2
## ..@ debug         : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.ticks        : <ggplot2::element_blank>
## $ axis.ticks.x      : NULL
## $ axis.ticks.x.top   : NULL
## $ axis.ticks.x.bottom : NULL
## $ axis.ticks.y      : NULL
## $ axis.ticks.y.left  : NULL
## $ axis.ticks.y.right : NULL
## $ axis.ticks.theta   : NULL
## $ axis.ticks.r       : NULL
## $ axis.minor.ticks.x.top : NULL

```

```

## $ axis.minor.ticks.x.bottom      : NULL
## $ axis.minor.ticks.y.left        : NULL
## $ axis.minor.ticks.y.right       : NULL
## $ axis.minor.ticks.theta         : NULL
## $ axis.minor.ticks.r             : NULL
## $ axis.ticks.length              : 'rel' num 0.5
## $ axis.ticks.length.x            : NULL
## $ axis.ticks.length.x.top        : NULL
## $ axis.ticks.length.x.bottom     : NULL
## $ axis.ticks.length.y           : NULL
## $ axis.ticks.length.y.left       : NULL
## $ axis.ticks.length.y.right      : NULL
## $ axis.ticks.length.theta        : NULL
## $ axis.ticks.length.r            : NULL
## $ axis.minor.ticks.length        : 'rel' num 0.75
## $ axis.minor.ticks.length.x      : NULL
## $ axis.minor.ticks.length.x.top  : NULL
## $ axis.minor.ticks.length.x.bottom : NULL
## $ axis.minor.ticks.length.y      : NULL
## $ axis.minor.ticks.length.y.left : NULL
## $ axis.minor.ticks.length.y.right : NULL
## $ axis.minor.ticks.length.theta  : NULL
## $ axis.minor.ticks.length.r      : NULL
## $ axis.line                      : <ggplot2::element_blank>
## $ axis.line.x                    : NULL
## $ axis.line.x.top                : NULL
## $ axis.line.x.bottom             : NULL
## $ axis.line.y                    : NULL
## $ axis.line.y.left               : NULL
## $ axis.line.y.right              : NULL
## $ axis.line.theta                : NULL
## $ axis.line.r                    : NULL
## $ legend.background              : <ggplot2::element_blank>
## $ legend.margin                  : NULL
## $ legend.spacing                 : 'rel' num 2
## $ legend.spacing.x              : NULL
## $ legend.spacing.y              : NULL
## $ legend.key                     : <ggplot2::element_blank>
## $ legend.key.size                : 'simpleUnit' num 1.2lines
##   .- attr(*, "unit")= int 3
## $ legend.key.height              : NULL
## $ legend.key.width               : NULL
## $ legend.key.spacing             : NULL
## $ legend.key.spacing.x          : NULL
## $ legend.key.spacing.y          : NULL
## $ legend.key.justification       : NULL
## $ legend.frame                   : NULL
## $ legend.ticks                   : NULL
## $ legend.ticks.length           : 'rel' num 0.2
## $ legend.axis.line              : NULL
## $ legend.text                    : <ggplot2::element_text>
##   .@ family      : NULL
##   .@ face        : NULL
##   .@ italic      : chr NA

```

```

## ..@ fontweight      : num NA
## ..@ fontwidth       : num NA
## ..@ colour          : NULL
## ..@ size             : 'rel' num 0.8
## ..@ hjust           : NULL
## ..@ vjust           : NULL
## ..@ angle           : NULL
## ..@ lineheight       : NULL
## ..@ margin          : NULL
## ..@ debug           : NULL
## ..@ inherit.blank: logi TRUE
## $ legend.text.position : NULL
## $ legend.title         : <ggplot2::element_text>
## ..@ family           : NULL
## ..@ face             : NULL
## ..@ italic           : chr NA
## ..@ fontweight       : num NA
## ..@ fontwidth        : num NA
## ..@ colour           : NULL
## ..@ size             : NULL
## ..@ hjust            : num 0
## ..@ vjust            : NULL
## ..@ angle            : NULL
## ..@ lineheight       : NULL
## ..@ margin          : NULL
## ..@ debug           : NULL
## ..@ inherit.blank: logi TRUE
## $ legend.title.position : NULL
## $ legend.position       : chr "right"
## $ legend.position.inside : NULL
## $ legend.direction      : NULL
## $ legend.byrow          : NULL
## $ legend.justification  : chr "center"
## $ legend.justification.top : NULL
## $ legend.justification.bottom : NULL
## $ legend.justification.left : NULL
## $ legend.justification.right : NULL
## $ legend.justification.inside : NULL
## [list output truncated]
## @ complete: logi TRUE
## @ validate: logi TRUE

```

#As we can see in this graph, almost no Kazakhstani citizens returned to their home country between 2020 and 2024. Although almost the same number of citizens from Azerbaijan and Uzbekistan emigrated to Türkiye, more Uzbekistani citizens returned to their home country.